Supplemental Results

Section I. Comparison to smaller cline datasets.

Supplemental Table 1. Table of p-values for comparison of lab loci to Teeter et al 2010 geographic clines widths.

	Bavarian Transect		Saxon Transect	
Lab Loci	All Chromosomes	Autosomes Only	All Chromosomes	Autosomes Only
Main Subset	0.006551*N	0.551641	0.748076	0.886604
Only Single QTL	0.121929	0.855682	0.64985	0.853559
Only Multiple QTL	0.014468*N	0.937376	0.769993	0.836052

NSites overlapping QTL have significantly narrower cline widths.

Supplemental Table 2. Table of p-values for comparison of lab loci to Gompert and Buerkle 2011 genomic clines.

	Bavarian Transect		Saxon Transect	
Lab Loci	All Chromosomes	Autosomes Only	All Chromosomes	Autosomes Only
Main Subset	0.203405	0.263211	0.166474	0.490041
Only Single QTL	0.696577	0.843653	0.173882	0.644223
Only Multiple QTL	0.35781	0.48381	0.127581	0.403412

Section II. Prezygotic Barriers

Supplemental Table 3. Prezygotic Isolation Papers.

Study	Source	Mapping Population	Genomic Coverage	Locus Type	Phenotype(s)
Laukaitis, Critser, and Karn 1997	Lab	C3H-AbpaB congenics	1 locus	Gene	Assortative mating based on androgen binding protein (ABP)
Loire et al. 2017	Nature	Danish Transect	RNAseq	Genes	Assortative mating
Smadja et al. 2015	Nature	Denmark	1,248 microsatellite loci	Microsatellite loci	NA

Supplemental Table 4. Results of permutation tests when prezygotic loci are included.

Data Subset		Treatment of QTL	
Loci	Chromosomes	Full QTL Intervals	1 Mb QTL Intervals
Main Dataset	All	0.4383	0.1241
	Autosomes Only	0.6913	0.1716
Including GWAS intervals	All	0.5545	0.1774
	Autosomes Only	0.6673	0.1439
Removing papers with a dual lab/nature approach	All	0.4785	0.8747
	Autosomes Only	0.4147	0.2430
Removing papers with lower confidence position conversions	All	0.4333	0.1707
	Autosomes Only	0.6895	0.1731

Supplemental Table 5. Comparison of prezygotic loci to cline widths.

Data Subset	Bavarian Transect		Czech Transect	
Loci	All Chromosomes	Autosomes Only	All Chromosomes	Autosomes Only
Assortative Mating Loci	0.379	0.568	0.533	0.448

Section III. Overlap within datasets.

It is difficult to make formal comparisons within our “lab” and “nature” datasets because the studies within each set are not independent of each other. In many cases, studies use the same sets of crosses or sample collections to look at new phenotypes or attributes. For example, Schwahn et al. (2018) looks at new phenotypes in the cross performed by White et al (2011). In some cases, subdividing the data too far removes power to detect excess overlap, either because the possible number of overlaps is too low, or there is too much asymmetry in the size of the two datasets. In our main dataset, we avoided this conflict by collapsing the datasets together (categorizing parts of the genome as hit by no studies vs one or more studies).

However, it is still important for interpreting our broader results to recognize that there is overlap within each dataset. We include the comparisons below to demonstrate this point.

Supplemental Table 6. Summary of comparisons within datasets.

Dataset 1	Dataset 2	Treatment	p-value
Lab-discovered candidate loci (QTL)	Lab-discovered genes (mainly from expression studies)	10,000 permutations using full QTL intervals	<2e-16
		10,000 permutations using 1Mb QTL intervals	<2e-16
Wang et al. (2011) cline widths from Bavarian transect	Nature loci used in the permutation tests (autosomes only)	Comparing cline widths at loci that do or do not overlap with dataset 2	<2e-16
Wang et al. (2011) cline widths from Czech transect	Nature loci used in the permutation tests (autosomes only)	Comparing cline widths at loci that do or do not overlap with dataset 2	<2e-16
Wang et al. (2011) cline widths from Czech transect	Wang et al. (2011) cline widths from Bavarian transect	Correlation Test (correlation=0.34)	<2e-16
Teeter et al (2008) cline widths from Saxony transect	Teeter et al. (2010) cline widths from Bavarian transect	Correlation Test (correlation=0.38)	0.0203

Supplemental Figure 1. Output of 10,000 permutations comparing lab-discovered genes to lab-discovered candidate loci (p-value=0).